Eulipotyphlan Molar from the Miocene Bukpyeong Formation, Donghae City, South Korea

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Introduction

- Eulipotyphlans are the third-largest group of mammals, comprising more than 500 modern species in four major groups: Erinaceidae (hedgehogs), Talpidae (moles), Soricidae (shrews), and Solenodontidae (solenodons) (Woodman, 2019), however, their Tertiary fossil record has not been previously documented in South Korea.
- The specimen (SNUVP 202501) was recovered in 2004 as part of microvertebrate fossils collected from the Miocene Bukpyeong Formation, through wet screening of fossiliferous sediments at an outcrop in Jiga-dong, Donghae City, South Korea with mammalian specimens *Spermophilinus* sp., *Democricetodon* or *Kowalskia* sp., and *Neocometes* aff. similis (Lee, 2004; Lee and Jacobs, 2010).
- Initially, the specimen was identified as a chiropteran molar, but it is here reidentified as a eulipotyphlan molar based on diagnostic dental features.
- This revised identification expands our understanding of Miocene mammalian diversity in Korea. Ongoing morphological analysis will aim to clarify its precise taxonomic placement within Eulipotyphla.

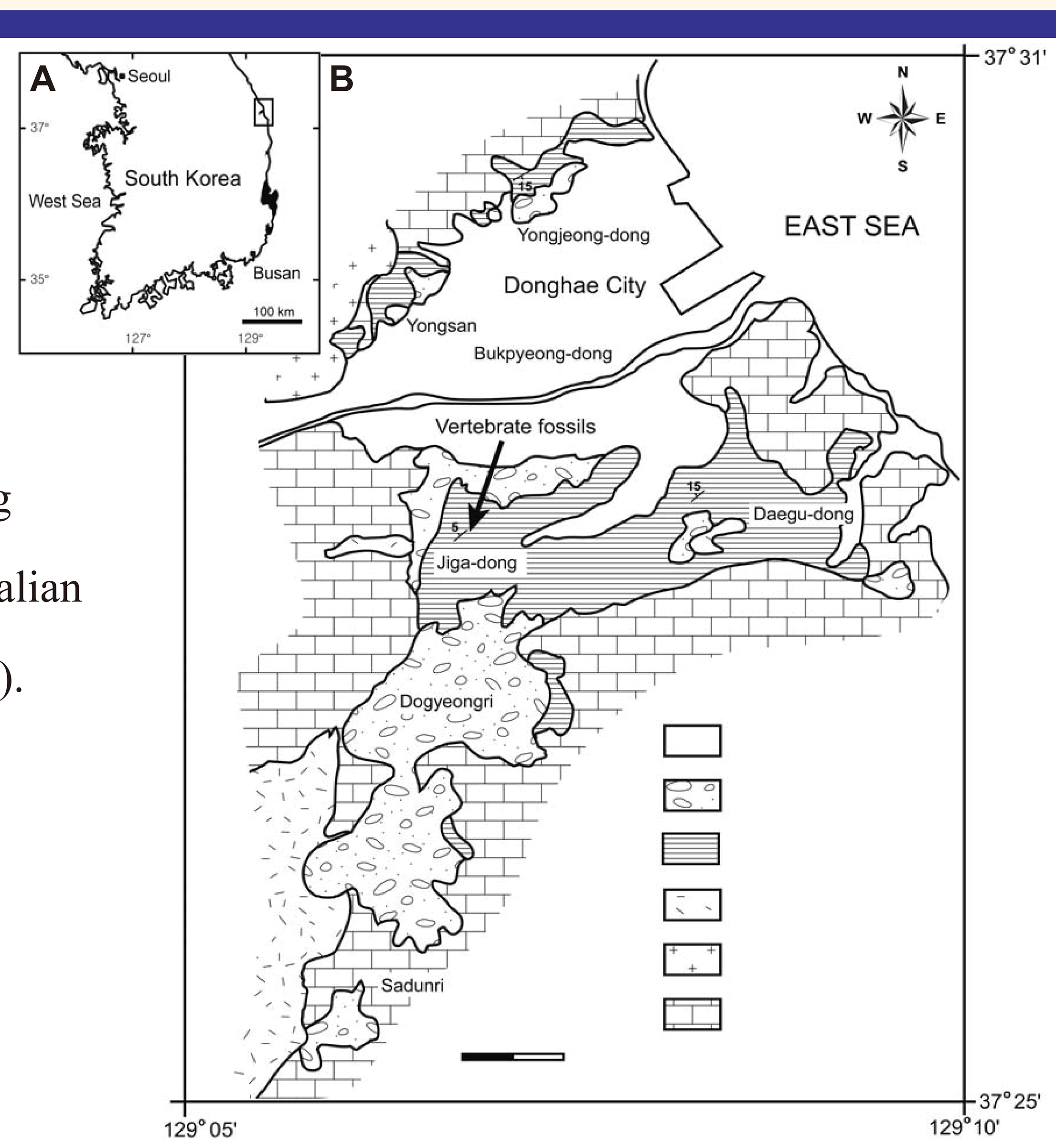

Fig. 1. Location of Tertiary basins in South Korea (A) and geologic map of the Bukpyeong area (B), (1) Yangnam Basin, (2) Pohang Basin, (3) Yeonghae Basin, (4) Bukpyeong Basin.

Description

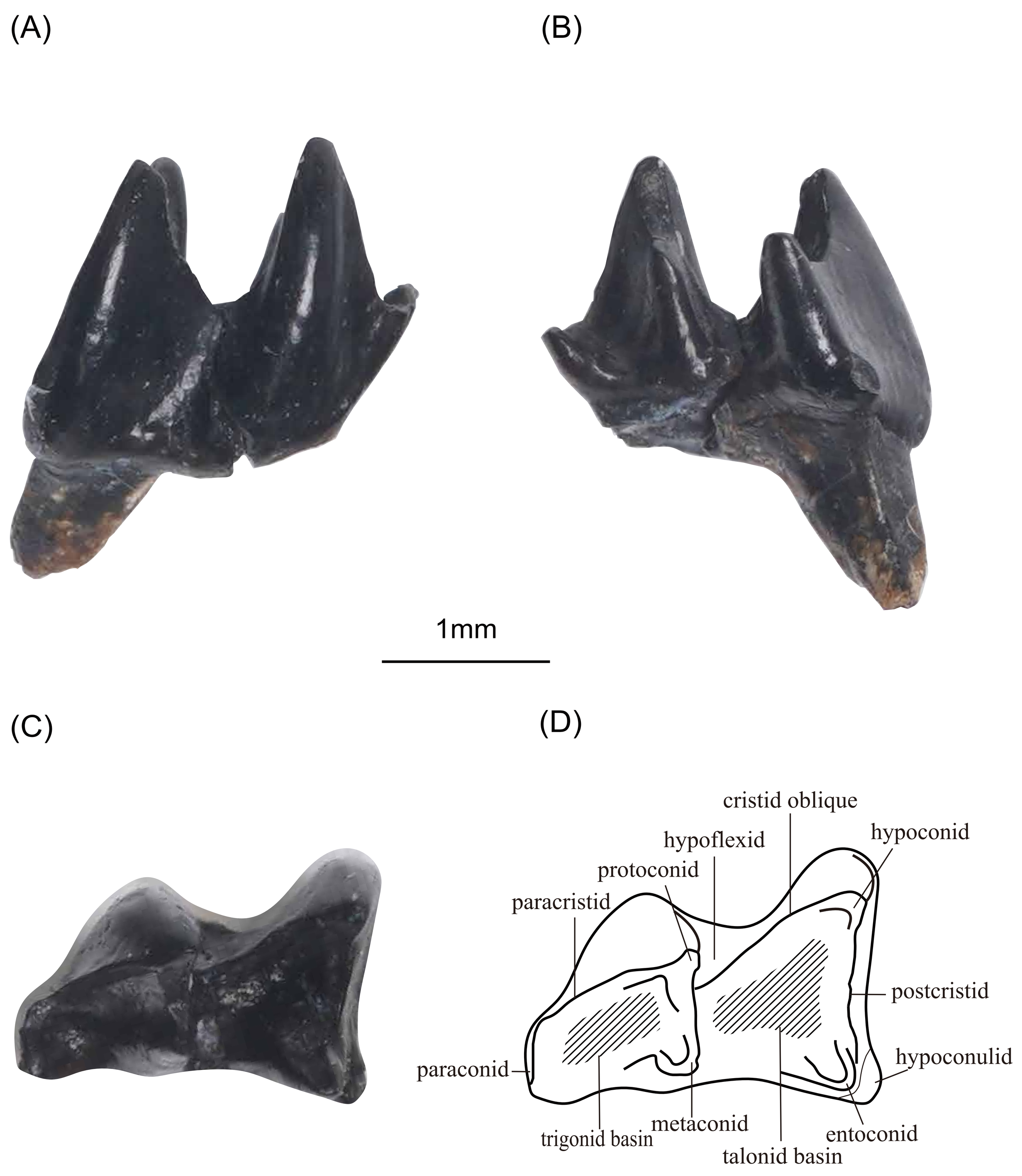

Fig. 2. Eulipotyphlan from the Miocene Bukpyeong Formation, South Korea. Right lower molar (SNUVP 202510) in labial (A) lingual (B) occlusal (C) views and terminology (D).

General tooth architecture and measurements

- Right lower molar (m1 or m2) with minimal wear, exhibiting insectivorous mammalian characteristics.
- Exodaenodont condition present, with the crown canted labially and overhangs the jawbone relative to the roots.
- Length: 2.11 mm, Trigonid width: 1.22 mm, Talonid width: 1.54 mm

Trigonid and talonid morphology

- U-shaped trigonid basin is broadly open lingually, with paracristid connecting paraconid and protoconid.
- Talonid is notably wider and open lingually, deep hypoflexid present between the protoconid and hypoconid.

Cusp height relationships and configuration

- All main cusps are very high and narrow, labial cusps (protoconid and hypoconid) are notably higher than lingual cusps.
- Among lingual cusps, the entoconid is the tallest, followed by the metaconid, and the low but distinct cuspate paraconid.
- A small hypoconulid (possibly broken) is positioned lingually, low behind the entoconid.

Cristid patterns and connections

- Postcristid extends from hypoconid directly to entoconid, while cristid oblique meets the posterior wall of trigonid centrally.
- No visible distal metacristid or entocristid present.

Cingulum structure and dental compression

- Lingual cingulum, precingulid, and postcingulid are absent, with only a small shelf at the hypoflexid base.
- Both talonid and trigonid are anteroposteriorly compressed, creating a narrow angle between cristid oblique and postcristid.

Discussion

Comparative Analysis

The absence of a clear buccal cingulum excludes this specimen from bat identification due to Miocene age, as cingulum presence is considered a synapomorphy of the crown group Chiroptera, dating back to the Eocene (Hand et al., 1994). The talonid is also distinctly wider than the trigonid, unlike typical bats where widths of talonid and trigonid are similar in m1 or m2.

The specimen differs from soricid molars, in that the entoconid is lower than the hypoconid, the postcristid connects the hypoconid to the entoconid, and from erinaceid molars in that the entoconid is lower than the hypoconid, hypoconulid located behind the entoconid (Tejedor et al., 2005).

Characteristics are most consistent with talpid molars, though further study is needed.

First Korean Peninsula record

It represents the first reported Eulipotyphla from the Miocene Bukpyeong Formation, extending the known geographic range of this order in East Asian Miocene deposits.

Future research directions

Further morphological study will focus on clarifying the specimen's taxonomic placement at the family level or lower within Eulipotyphla.

Comparative studies with Miocene eulipotyphlan fossils from Asia and Europe will be essential to clarify phylogeny and biogeographic implications.

References

- Hand, S. J., Novacek, M., Godthelp, H., & Archer, M. (1994). First Eocene bat from Australia. Journal of Vertebrate Paleontology, 14(3), 375-381.
- Lee, Y. N., & Jacobs, L. L. (2010). The platacanthomyine rodent Neocometes from the Miocene of South Korea and its paleobiogeographical implications. Acta Palaeontologica Polonica, 55(4), 581-586.
- Lee, Y. N. (2004). The first cyprinid fish and small mammal fossils from the Korean Peninsula. Journal of Vertebrate Paleontology 24(2), 489-493.
- Tejedor, M. F., Czaplewski, N. J., Goin, F. J., & Aragón, E. (2005). The oldest record of South American bats. Journal of Vertebrate Paleontology, 25(4), 990-993.
- Woodman, N. (2019). American recent Eulipotyphla: Nesophontids, solenodons, moles, and shrews in the New World. Smithsonian Institution Scholarly Press, 1-3.